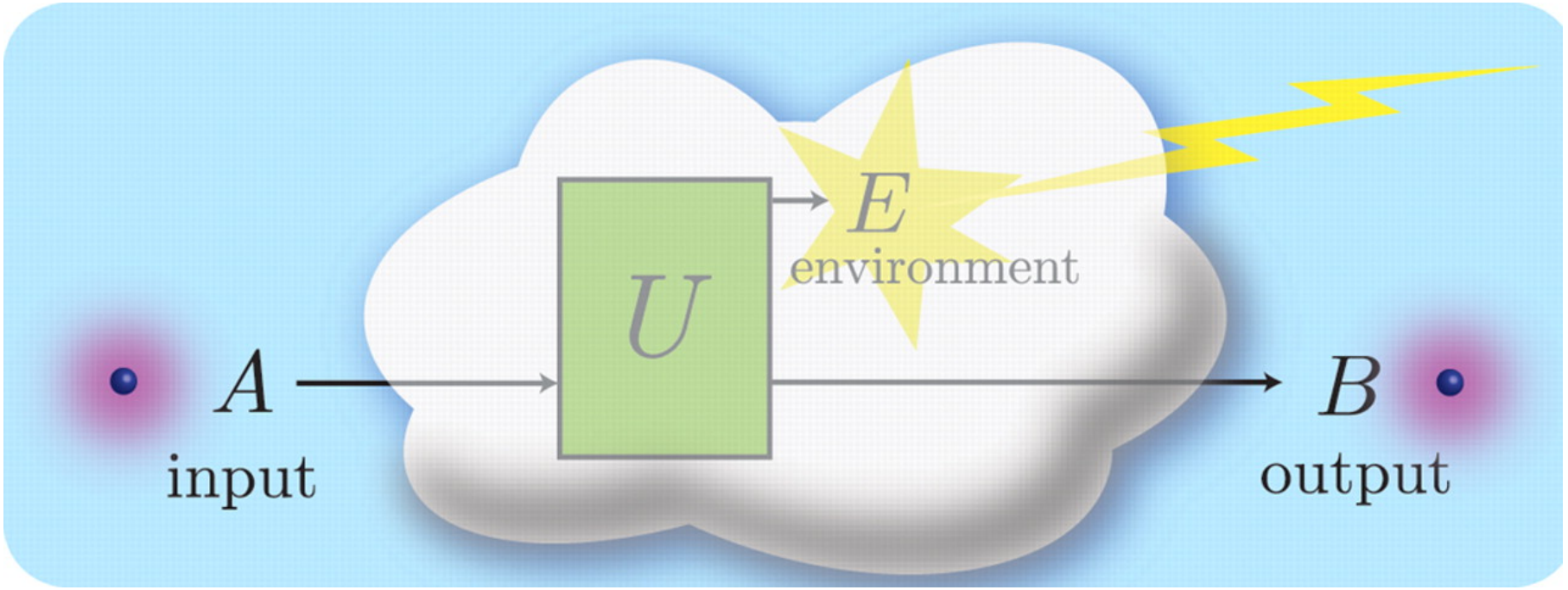
PHYS7045 – Quantum Channels

+ Thermal states



Nielsen & Chuang, Chapter 8

Properties of density matrices:

$$\hat{\rho} = \begin{bmatrix} \cdots & \rho_{ij} & \cdots \\ & \vdots & \end{bmatrix}_{n \times n} = \sum_{ij} \rho_{ij} |i\rangle\langle j| \qquad \text{where } \dim \mathcal{H}_S = n$$

(=) $\rho_{ij} = \langle i | \hat{\rho} | j \rangle$

- $\hat{\rho}$ is Hermitian $\rho_{ij} = \rho_{ji}^*$ so that e/vals are real
  ⇒ some basis that diagonalizes $\hat{\rho} = \sum_k \rho_{kk} |k\rangle\langle k|$ in some basis $\{|k\rangle\}$
- for a pure state $|0\rangle$: $\rho_0 = |0\rangle\langle 0| \Rightarrow \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ 0 & \cdots & & \\ \vdots & & & \end{bmatrix}$ $\{|0\rangle, |1\rangle \ldots\}$
  has e/vals 1 + 0 (many copies)
- $\underline{\mathrm{tr}\,\rho_0 = 1}$ for all density matrices
- ρ is "positive semi-definite" ⇒ $0 \le \rho_{kk} \le 1$ (non-negative e/vals)

$\Rightarrow$ interpret eigenvals as probabilities for different outputted states

$$\rho = p\,|0\rangle\langle 0| + (1-p)\,|1\rangle\langle 1| = \begin{pmatrix} p & 0 \\ 0 & 1-p \end{pmatrix}$$

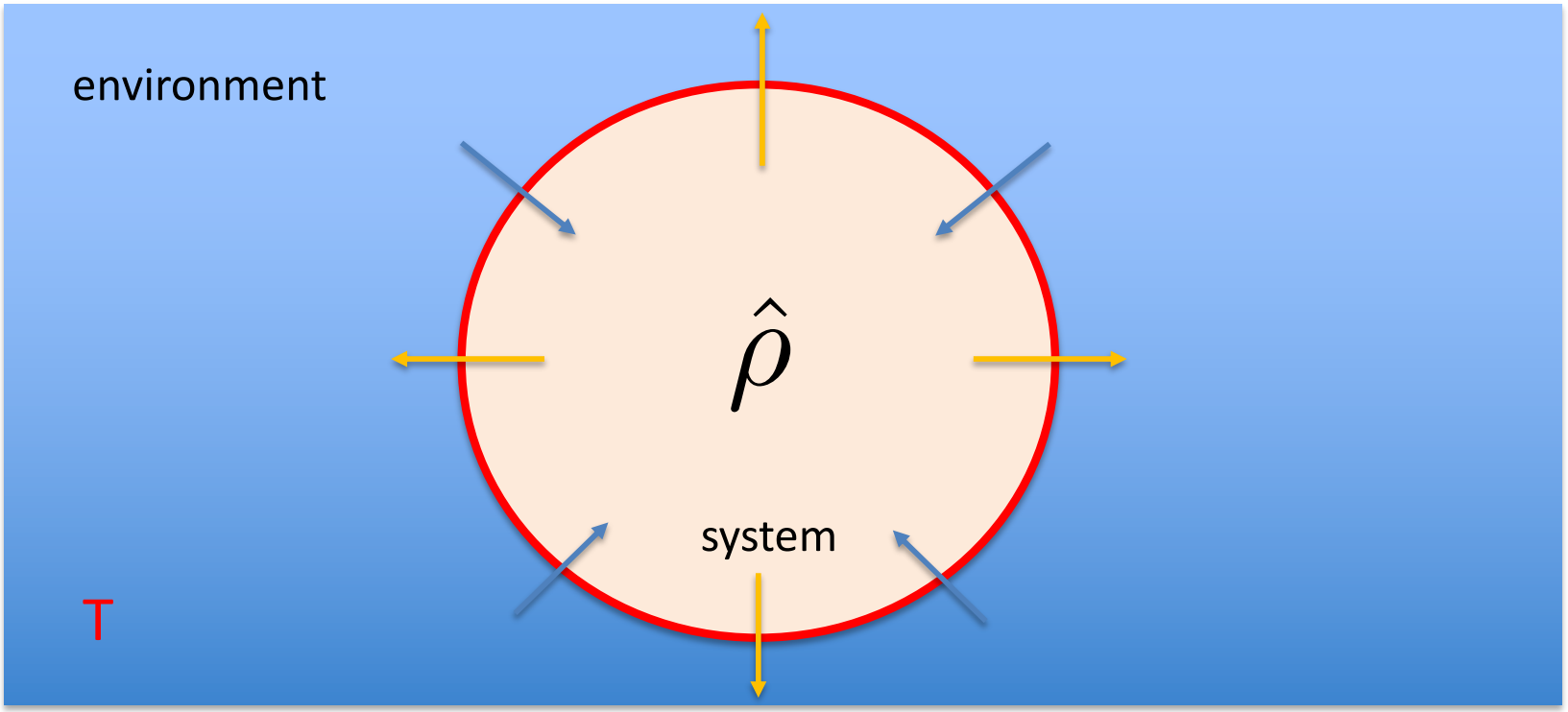
in general we can diagonalize $\hat{\rho}$:

$$\hat{\rho} = \begin{bmatrix} \rho_{00} & 0 & \\ 0 & \rho_{11} & \\ & & \rho_{22} \\ & & & \ddots \end{bmatrix} = \begin{bmatrix} \rho_0 & 0 & \\ 0 & \rho_1 & \\ & & \rho_2 \\ & & & \ddots \end{bmatrix}$$

Simple models of Open Quantum Systems

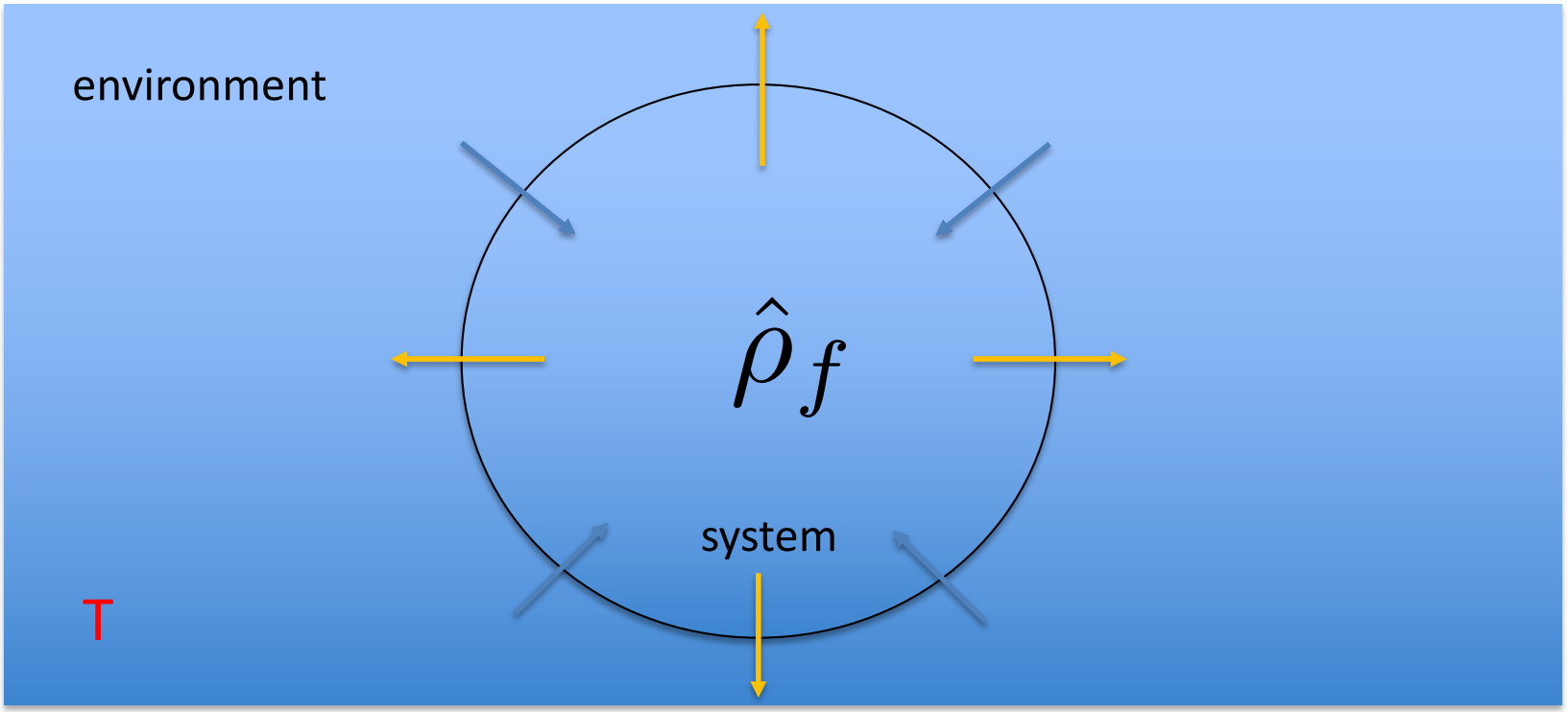
- The density operator of a system in thermal equilibrium
- Quantum channels
- Quantum noise and quantum decoherence in simple systems

Thermal equilibrium



- A quantum system interacts with the environment at temperature T…

Thermal equilibrium



- A quantum system interacts with the environment at temperature T…
- …and comes into thermal equilibrium with it

⇒ system will equilibrate to temperature T

What is the state of a classical system with system energy $E_0, E_1, \ldots$, at temp T?

E_n
—⋮
E_2
E_1
E_0

Maxwell Boltzmann:

Prob (of finding the system in classical state j)

$\propto e^{-E_j/kT}$

$= \frac{e^{-E_j/kT}}{Z}$

$Z = \sum_j e^{-E_j/kT}$ partition function.

a.k.a "classical" → density matrix

$= \frac{1}{Z}\begin{bmatrix} e^{-E_0/kT} & 0 & 0 & \\ 0 & e^{-E_1/kT} & 0 & \\ 0 & 0 & e^{-E_2/kT} & \ddots \end{bmatrix}$

as $T \to 0$,

$$Prob(j=0) \to 1$$

$$Prob(j>0) \to 0$$

$T \to \infty$

$$Prob(j) \to const$$

$$P^{(collapsed)} = \begin{bmatrix} 1/n & & 0 \\ & 1/n & \\ 0 & & \ddots \\ & & & 1/n \end{bmatrix}_{n \times n} = \frac{1}{n} \mathbb{1}$$

Thermal equilibrium

- Von Neuman entropy of a quantum state is

$$S = -Tr\{\hat{\rho}\, log\hat{\rho}\}$$

$$= -\sum_{k}^{N} \rho_{kk}^{(diag)}\ log\rho_{kk}^{(diag)}$$

where the 2nd line is expressed in the diagonal representation and *N* is the dimension of Hilbert space

- It is zero for a pure state
- For a completely random state it has a maximum value of

$$S = logN$$

Thermal equilibrium

- We can identify the density operator for thermal equilibrium by maximizing the entropy under an energy constraint
- In equilibrium $\dot{\hat{\rho}} = 0$ so $[\hat{\rho}, \hat{H}] = 0$ and they can be simultaneously diagonalized such that

$$\hat{\rho} = \sum_k \rho_{kk} |E_k\rangle\langle E_k| \quad \hat{H} = \sum_k E_k |E_k\rangle\langle E_k| \quad \Rightarrow \quad \langle \hat{H} \rangle = \sum_k \rho_{kk} E_k$$

$$= \mathrm{tr}(\hat{H}\hat{\rho})$$

von Neumann: $\dot{\hat{\rho}} = -\frac{i}{\hbar}[\hat{H}, \hat{\rho}] = 0$

- Hence, we wish to find the $\hat{\rho}$ such that $\delta S = 0$ under the constraints

$$\delta \langle \hat{H} \rangle = 0 \qquad \delta Tr\{\hat{\rho}\} = 0$$

(fixed energy) (normalization)

Thermal equilibrium

- By the method of Lagrange multipliers we have

$$\sum_k \delta\rho_{kk}(log\rho_{kk} + 1 + \beta E_k + \gamma) = 0$$

leading (after normalization) to

$$\rho_{kk} = \frac{e^{-\beta E_k}}{\sum_l e^{-\beta E_l}}$$

or

$$\hat{\rho} = \frac{e^{-\beta \hat{H}}}{Tr\{e^{-\beta \hat{H}}\}}$$

where $\beta = \frac{1}{kT}$

Constrained optimisation of S, form a Lagrange syst

$$U = S + \beta \langle H \rangle + \gamma \, \mathrm{tr}(\rho)$$

β, γ: Lagrange multipliers

$\langle H \rangle = kT$

$\mathrm{tr}(\rho) = 1$

$$= \sum_k \rho_{kk} \log \rho_{kk} + \beta E_k \rho_{kk} + \gamma \rho_{kk}$$

extremise this const func.

$$\frac{\partial U}{\partial \rho_{kk}} = \log \rho_{kk} + 1 + \beta E_k + \gamma = 0 \quad \text{for all } \rho_{kk}$$

$$\rho_{kk} = e^{-\gamma - \beta E_k - 1}$$

$$\mathrm{tr}\,\rho = \sum_k \rho_{kk} = 1 \Rightarrow \sum_k e^{-\gamma - \beta E_k - 1} = 1$$

$$= e^{-\gamma - 1} \sum_k e^{-\beta E_k} \Rightarrow e^{-\gamma - 1} = \frac{1}{\sum_k e^{-\beta E_k}}$$

$$\Rightarrow \rho_{kk} = \frac{e^{-\beta E_k}}{\sum_k e^{-\beta E_k}} = \frac{e^{-\beta \hat{H}}}{\mathrm{tr}\, e^{-\beta \hat{H}}} \equiv \frac{e^{-\beta \hat{H}}}{Z}$$

$$H = \begin{bmatrix} E_0 & & & 0 \\ & E_1 & & \\ & & E_2 & \\ 0 & & & \ddots \end{bmatrix}$$ in the energy eigen basis's

$$\rho_{thermal} = \frac{e^{-\beta H}}{Z} = \frac{1}{Z}\begin{bmatrix} e^{-\beta E_0} & & & 0 \\ & e^{-\beta E_1} & & \\ & & e^{-\beta E_2} & \\ 0 & & & \ddots \end{bmatrix}$$

this is at least a good candidate for the ground thermal state

Thermalisation with an environment at temp T takes any initial state to the thermal state of the system

$$\rho_{th} = \frac{e^{-\beta \hat{H}}}{\mathrm{tr}(e^{-\beta \hat{H}})}$$

- This is the steady state, & the evolution from an initial state depends on the details of the interaction between the Sys & its environment.

Evolution of a closed system:

von Neumann $\dot{\rho} = -i[H, \rho]$ $\Leftrightarrow$ $|\dot{\psi}\rangle = -iH|\psi\rangle$

$\Rightarrow |\psi(t)\rangle = \underbrace{e^{-iHt}}_{U(t)} |\psi(0)\rangle$

$\Rightarrow \rho(0) \xrightarrow{\text{channel}} \rho(t) = |\psi(t)\rangle\langle\psi(t)|$ (if ρ is pure)

$= U(t)|\psi(0)\rangle\langle\psi(0)|U^\dagger(t)$

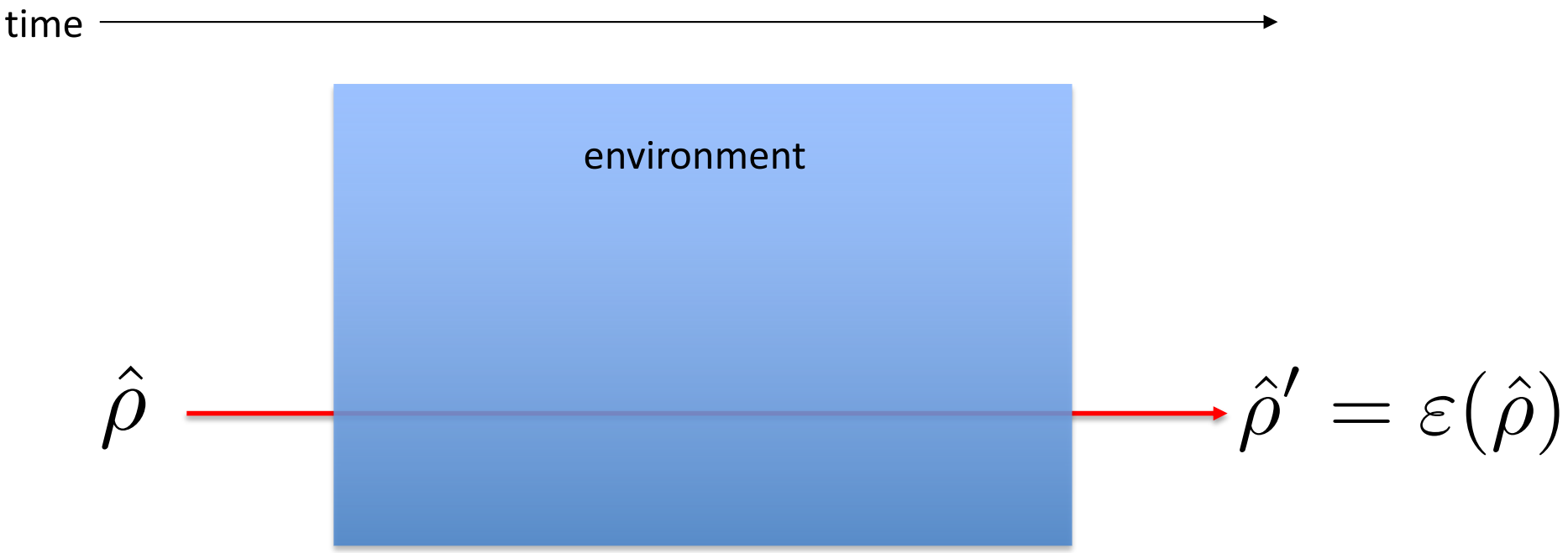
is also true for any initial mixture of pure states,

$$\rho(t) = U(t)\rho(0)U^\dagger(t)$$

— "Unitary Quantum Channel"

— this is a special case of general Quantum Channels

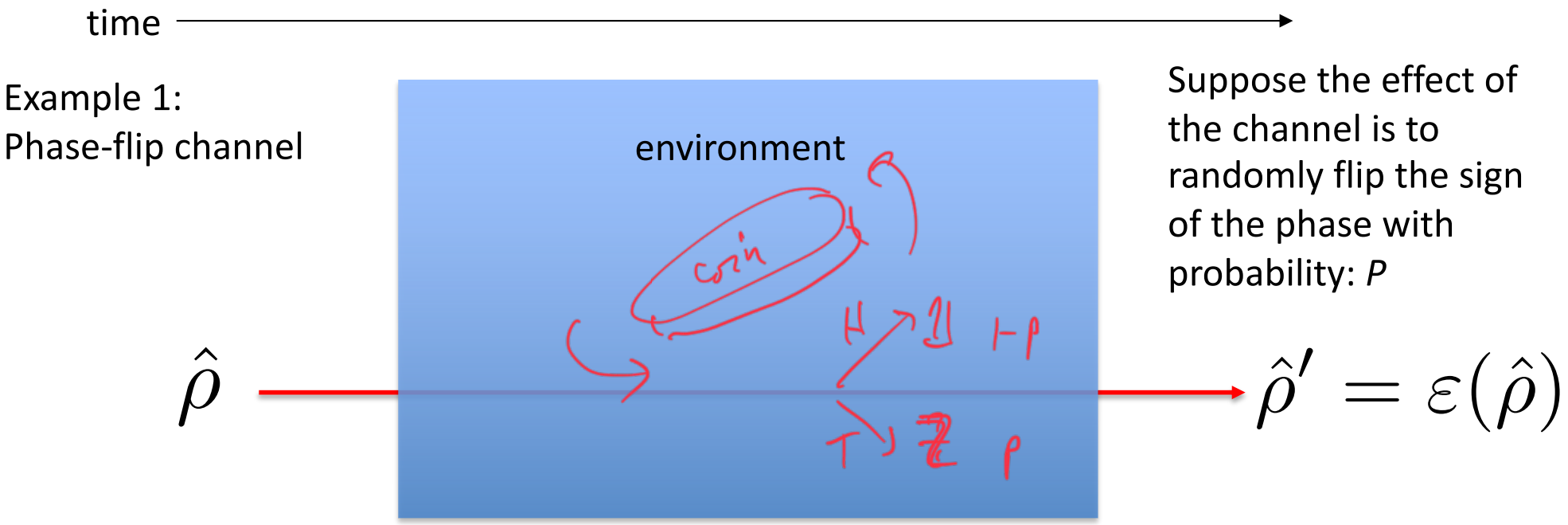
Quantum Channels



- Consider a quantum system which is isolated, interacts with the environment for a certain time, then is isolated again

eg
- unitary evolution : $\rho' = \mathcal{E}(\rho) = U \rho U^\dagger$
- thermalisation : $\rho' = \mathcal{E}(\rho) = \frac{e^{-\beta H}}{\mathrm{tr}\, e^{-\beta H}}$

Quantum Noise



- Consider a quantum system which is isolated, interacts with the environment for a certain time, then is isolated again

If Coin is Heads $\rho \to \rho_H' = \mathbb{1}\rho\mathbb{1}$ with prob $1-p$

" " fails $\rho \to \rho_T' = Z\rho Z^\dagger$ " p

Calculate $\rho' = p\,\rho_H' + (1-p)\,\rho_T'$ $\quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ in $\{|0\rangle, |1\rangle\}$ basis

Suppose input $\rho = |\psi(0)\rangle\langle\psi(0)|$ where $|\psi(0)\rangle = \mu|0\rangle + \nu|1\rangle$

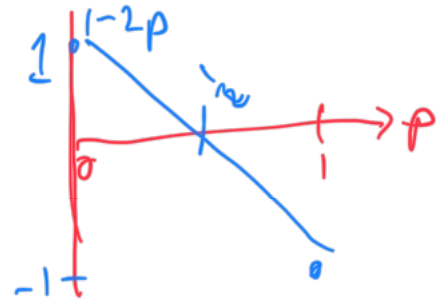
$$\rho_H' = \rho = |\psi(0)\rangle\langle\psi(0)| = (\mu|0\rangle + \nu|1\rangle)(\mu^*\langle 0| + \nu^*\langle 1|) = |\mu|^2|0\rangle\langle 0| + \mu\nu^*|0\rangle\langle 1| + \mu^*\nu|1\rangle\langle 0| + |\nu|^2|1\rangle\langle 1|$$

$$= \begin{bmatrix} |\mu|^2 & \mu\nu^* \\ \mu^*\nu & |\nu|^2 \end{bmatrix}$$

$$\rho_T' = Z\rho Z^\dagger = (Z|\psi(0)\rangle)(\langle\psi(0)|Z^\dagger) = \begin{bmatrix} |\mu|^2 & -\mu\nu^* \\ -\mu^*\nu & |\nu|^2 \end{bmatrix} \quad \text{since } Z|\psi(0)\rangle = \mu|0\rangle - \nu|1\rangle$$

$$\Rightarrow \rho' = \begin{bmatrix} p|\mu|^2 + (1-p)|\mu|^2 & -p\mu\nu^* + (1-p)\mu\nu^* \\ -p\mu^*\nu + (1-p)\mu^*\nu & p|\nu|^2 + (1-p)|\nu|^2 \end{bmatrix}$$

$$= \begin{bmatrix} |\mu|^2 & (1-2p)\mu\nu^* \\ (1-2p)\mu^*\nu & |\nu|^2 \end{bmatrix}$$



Quantum Noise

- Consider a quantum system which is isolated, interacts with the environment for a certain time, then is isolated again

$$\varepsilon(\hat{\rho}) = (1-P)(\mu|0\rangle + \nu|1\rangle)(\mu^*\langle 0| + \nu^*\langle 1|) + P(\mu|0\rangle - \nu|1\rangle)(\mu^*\langle 0| - \nu^*\langle 1|)$$

Expand:

Quantum Noise

- Consider a quantum system which is isolated, interacts with the environment for a certain time, then is isolated again

$$\varepsilon(\hat{\rho}) = (1-P)(\mu|0\rangle + \nu|1\rangle)(\mu^*\langle 0| + \nu^*\langle 1|) + P(\mu|0\rangle - \nu|1\rangle)(\mu^*\langle 0| - \nu^*\langle 1|)$$

$$= |\mu|^2|0\rangle\langle 0| + |\nu|^2|1\rangle\langle 1| + (1-2P)(\mu\nu^*|0\rangle\langle 1| + \mu^*\nu|1\rangle\langle 0|)$$

Quantum Decoherence

- Consider a quantum system which is isolated, interacts with the environment for a certain time, then is isolated again

$$\varepsilon(\hat{\rho}) = (1-P)(\mu|0\rangle + \nu|1\rangle)(\mu^*\langle 0| + \nu^*\langle 1|) + P(\mu|0\rangle - \nu|1\rangle)(\mu^*\langle 0| - \nu^*\langle 1|)$$

$$= |\mu|^2|0\rangle\langle 0| + |\nu|^2|1\rangle\langle 1| + (1-2P)(\mu\nu^*|0\rangle\langle 1| + \mu^*\nu|1\rangle\langle 0|)$$

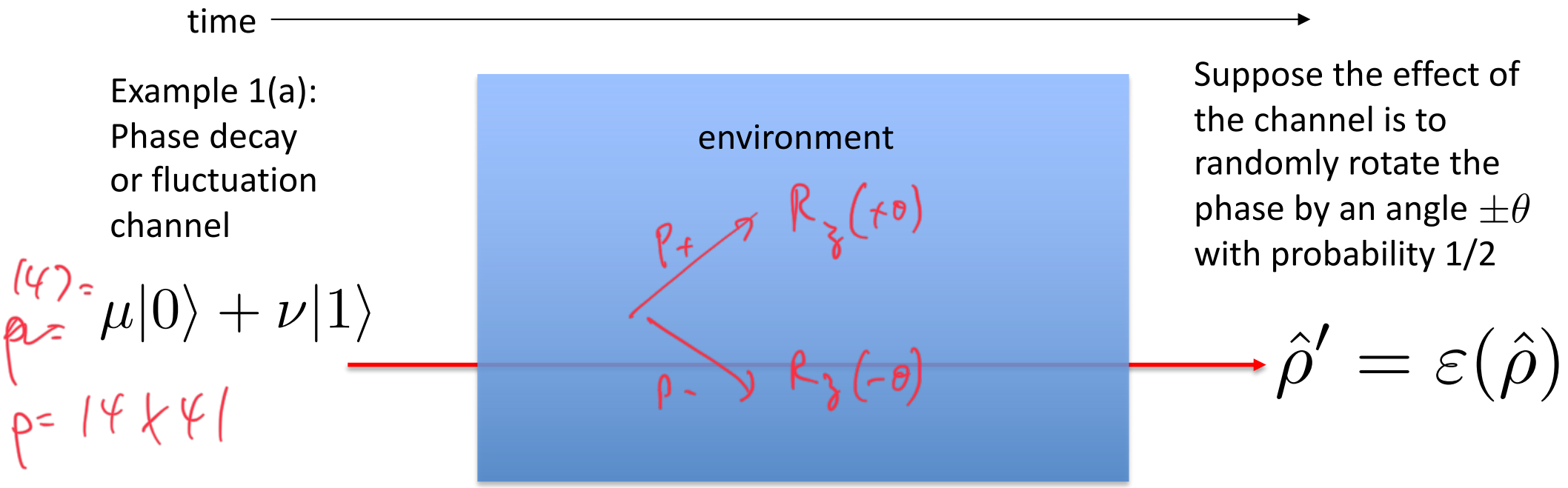
So the offdiagonal elements evolve under the prob. phase-flip channel

$$\mu\nu^* \Rightarrow \mu\nu^*(1-2p)$$

- A measurement of the final state in the computational basis still gives probs $|\mu|^2, |\nu|^2$ for outcomes $|0\rangle$ & $|1\rangle$

- A measurement in the x-basis $|\pm\rangle = \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}$ will have different probabilities before & after the channel.

Quantum Noise



- Consider a quantum system which is isolated, interacts with the environment for a certain time, then is isolated again

$\rho' = \varepsilon(\hat{\rho}) = p_+ R_z(+\theta)\,\rho\,R_z^\dagger(+\theta) + p_- R_z(-\theta)\,\rho\,R_z^\dagger(-\theta)$ $\quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

$$R_z(+\theta)|+\rangle = \mu|0\rangle + \nu e^{i\theta}|1\rangle$$
$$R_z(-\theta)|+\rangle = \mu|0\rangle + \nu e^{-i\theta}|1\rangle$$

$$R_z(\theta) = e^{-i\theta Z} = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta Z} \end{bmatrix}$$

$$\rho' = p_+ \begin{bmatrix} |\mu|^2 & e^{+i\theta}\mu\nu^* \\ e^{-i\theta}\mu^*\nu & |\nu|^2 \end{bmatrix} + p_- \begin{bmatrix} |\mu|^2 & e^{-i\theta}\mu\nu^* \\ e^{+i\theta}\mu^*\nu & |\nu|^2 \end{bmatrix}$$

$$= \begin{bmatrix} |\mu|^2 & \mu\nu^*(p_+e^{i\theta} + p_-e^{-i\theta}) \\ \mu^*\nu(p_+e^{-i\theta} + p_-e^{i\theta}) & |\nu|^2 \end{bmatrix}$$

$$R_z(\mp\theta)\rho R_z^\dagger(\mp\theta) = \begin{bmatrix} |\mu|^2 & e^{\mp i\theta}\mu\nu^* \\ e^{\mp i\theta}\mu^*\nu & |\nu|^2 \end{bmatrix}$$

if $p_+ = p_- = \frac{1}{2}$ then $\rho' = \begin{bmatrix} |\mu|^2 & \mu\nu^*\cos\theta \\ \mu^*\nu\cos\theta & |\nu|^2 \end{bmatrix}$ $+1 \le \cos\theta < 1$

Compare with phase-flip channel $\rho' = \begin{bmatrix} |\mu|^2 & (1-2p)\mu\nu^* \\ (1-2p)\mu^*\nu & |\nu|^2 \end{bmatrix}$ $+1 < 1-2p < 1$

So the phase-electric channel is equivalent to the phase flip channel

Quantum Noise

- Consider a quantum system which is isolated, interacts with the environment for a certain time, then is isolated again

$$\varepsilon(\hat{\rho}) = \frac{1}{2}(\mu|0\rangle + \nu|1\rangle)(\mu^*\langle 0| + \nu^*\langle 1|) +$$
$$\frac{1}{4}(\mu|0\rangle + e^{i\theta}\nu|1\rangle)(\mu^*\langle 0| + e^{-i\theta}\nu^*\langle 1|) + \frac{1}{4}(\mu|0\rangle + e^{-i\theta}\nu|1\rangle)(\mu^*\langle 0| + e^{i\theta}\nu^*\langle 1|)$$

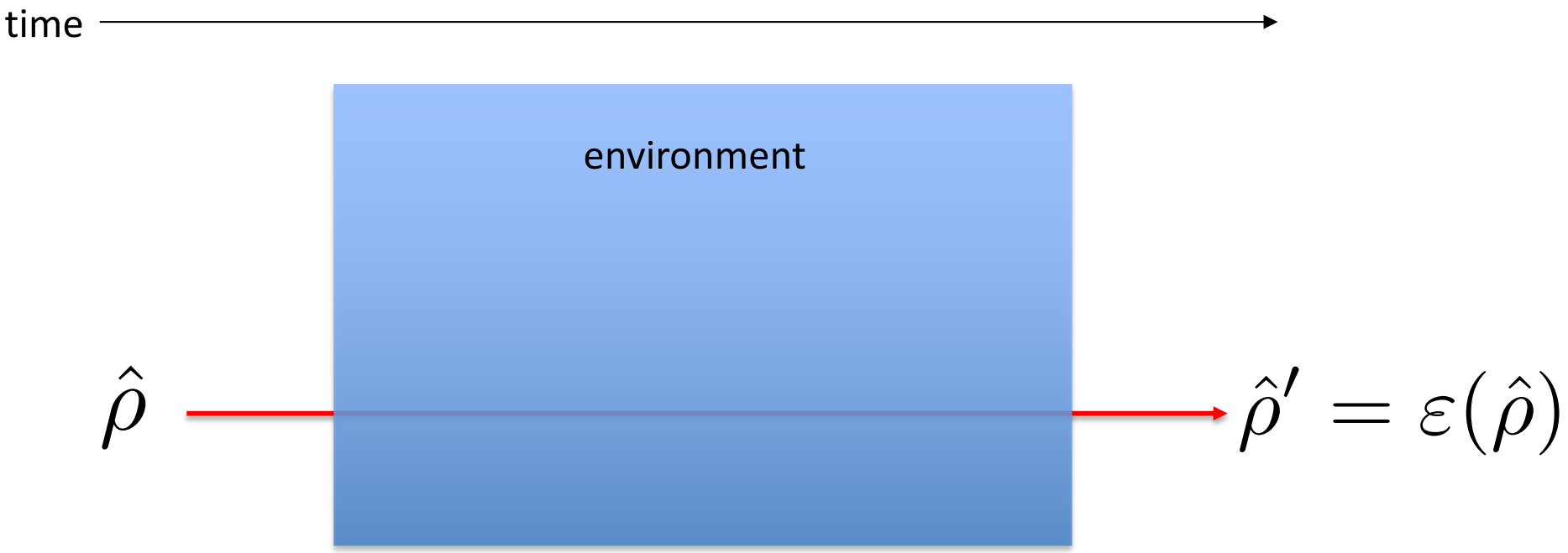
Quantum Decoherence

- Consider a quantum system which is isolated, interacts with the environment for a certain time, then is isolated again

$$\varepsilon(\hat{\rho}) = |\mu|^2|0\rangle\langle 0| + |\nu|^2|1\rangle\langle 1| + (1-2P)(\mu\nu^*|0\rangle\langle 1| + \mu^*\nu|1\rangle\langle 0|)$$

- Phase flip channel = Phase decay/fluctuation channel

Quantum Channels



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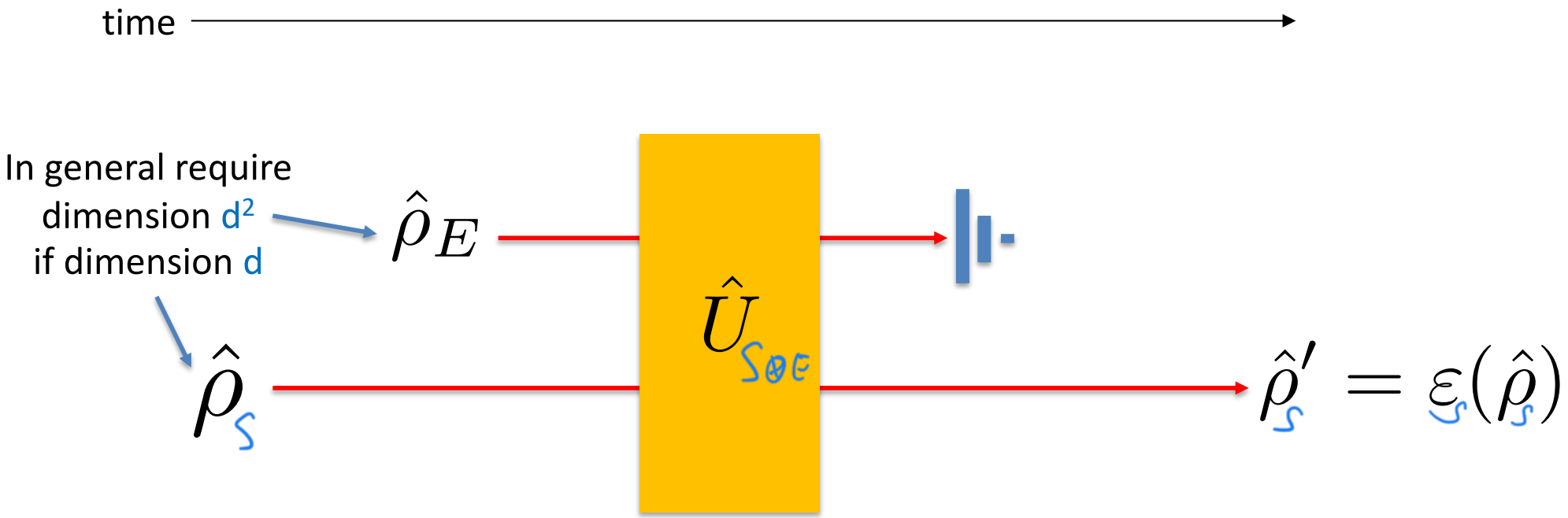
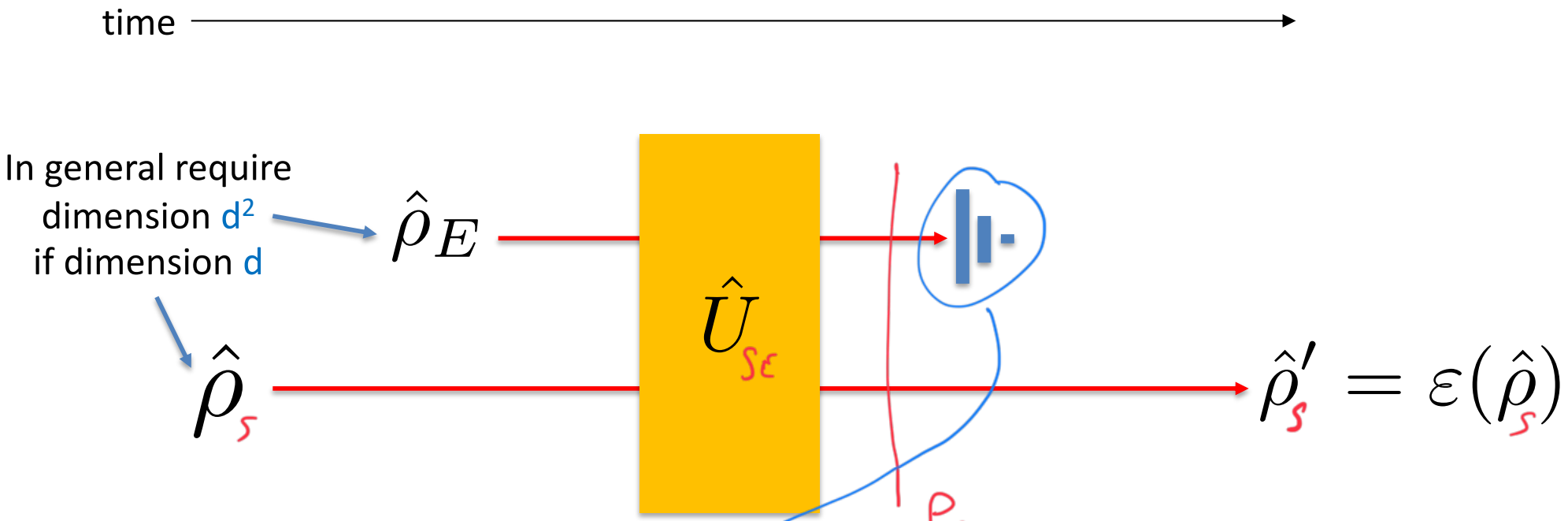
Quantum Channels



- Consider a quantum system which is isolated, interacts with the environment for a certain time, then is isolated again
- Sometimes this can be modelled as the interaction with a single "environment system" which is then traced out

Quantum Channels



$$\varepsilon_S(\hat{\rho}_S) = tr_E[\hat{U}_{SE}(\hat{\rho} \otimes \hat{\rho}_E)\hat{U}_{SE}^{\dagger}] \in \mathcal{H}_S$$

ρ_{SE}

tr_E : average over all configurations of the environment ⇔ we "forget" the state of the environment.

Initial separable state of system & environment

$$\rho_s' = \mathcal{E}(\rho_s) = \mathrm{tr}_E \left(U \rho_s \otimes \rho_e U^\dagger \right)$$

$$= \sum_{i=0}^{m} {}_E\langle e_i | \, U \rho_s \otimes \rho_e U^\dagger \, | e_i \rangle_E$$

$\mathcal{H}_s = \mathrm{span}\{|0\rangle_s, |1\rangle_1, \ldots\}_n$

$\mathcal{H}_E = \mathrm{span}\{|e_0\rangle, |e_1\rangle, \ldots\}_m$

$n \leq m$

$\rho_s = [\quad]_{n \times n} \in \mathcal{H}_s$, $\rho_e = [\quad]_{m \times m} \in \mathcal{H}_E$

$U \rho_s \otimes \rho_e U^\dagger = [\quad]_{(nm) \times (nm)} \in \mathcal{H}_s \otimes \mathcal{H}_E$

$\rho_s' = \sum_{i=1}^{m} \ldots = [\quad]_{n \times n} \in \mathcal{H}_s$

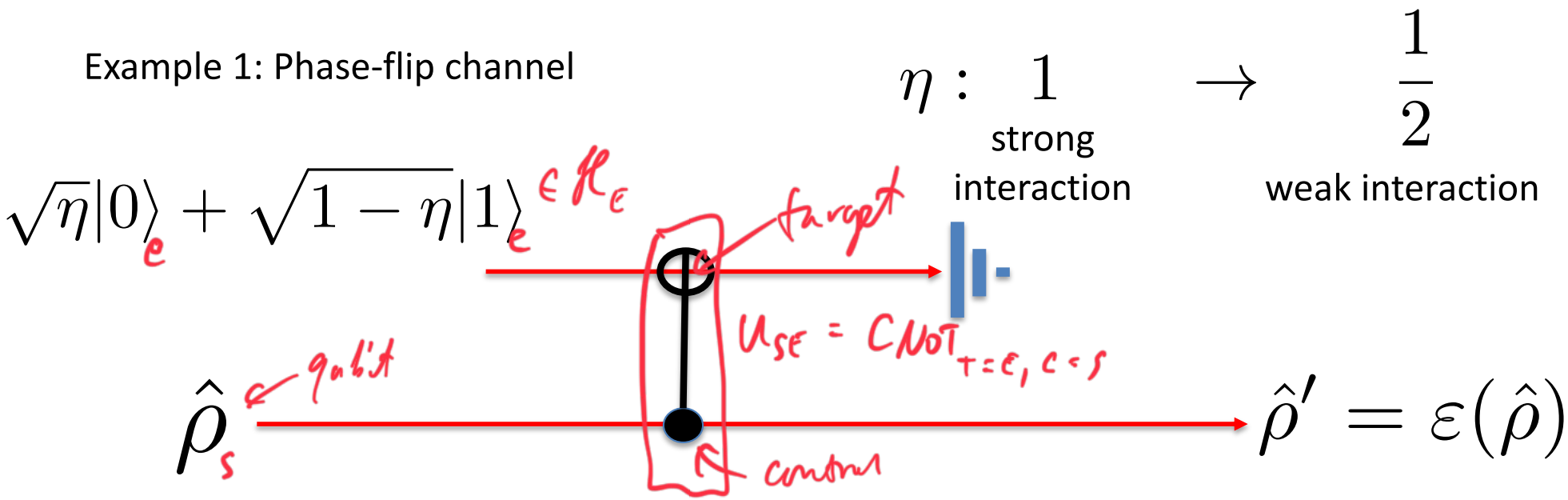
Quantum Channels

Example 1: Phase-flip channel

$$\eta : \quad 1 \quad \rightarrow \quad \frac{1}{2}$$

strong interaction

weak interaction



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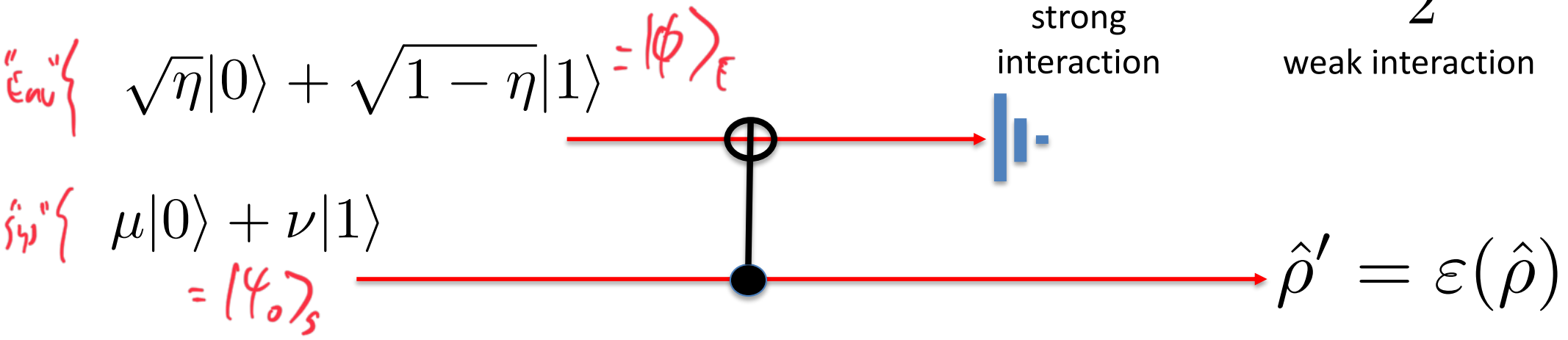


$$\varepsilon(\hat{\rho}_S) = \quad \mathrm{tr}_E \left(U_{SE} \, \rho_S \otimes \rho_E \, U_{SE}^{\dagger} \right) = \rho_S'$$

$$= \mathrm{tr}_E \left(C_{NOT} \, |\psi_0\rangle_S\langle\psi_0| \otimes |\phi\rangle_E\langle\phi| \, C_{NOT}^{\dagger} \right)$$

$$= \mathrm{tr}_E \left(\left[C_{NOT} \left(|\psi_0\rangle_S |\phi\rangle_E \right) \right] \left[\left({}_S\langle\psi_0| \otimes {}_E\langle\phi| \right) C_{NOT}^{\dagger} \right] \right)$$

CNOT $\quad \mathcal{H}_s = \mathrm{span}\{|0\rangle_s, |1\rangle_s\}$ $\quad \mathcal{H}_e = \mathrm{span}\{|0\rangle_e, |1\rangle_e\}$

$|0_c 0_T\rangle = |00\rangle$
$|0\ 1\rangle \to |01\rangle$
$|1\ 0\rangle = |11\rangle$
$|1\ 1\rangle = |10\rangle$

$\Leftrightarrow$ $\quad CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$

$$|\Psi_{input}\rangle_{se} = |\psi\rangle_s \otimes |\phi\rangle_e = (\mu|0\rangle_s + \nu|1\rangle_s) \otimes (\sqrt{n}\,|0\rangle_e + \sqrt{1-n}\,|1\rangle_e)$$

$$CNOT\,|\Psi_{input}\rangle_{se} = CNOT\left(\mu\sqrt{n}|00\rangle + \mu\sqrt{1-n}|01\rangle + \nu\sqrt{n}|10\rangle + \nu\sqrt{1-n}|11\rangle\right)$$

$$= \mu\sqrt{n}\,|00\rangle_{se} + \mu\sqrt{1-n}\,|01\rangle + \nu\sqrt{n}\,|11\rangle + \nu\sqrt{1-n}\,|10\rangle = |\Psi'\rangle_{se}$$

$$\langle\Psi'| = \mu^*\sqrt{n}\langle 00| + \mu^*\sqrt{1-n}\langle 01| + \nu^*\sqrt{n}\langle 11| + \nu^*\sqrt{1-n}\langle 10|$$

$$\rho' = tr_s\,|\Psi'\rangle\langle\Psi'| = {}_e\langle 0|\Psi'\rangle_{se}\,\langle\Psi'|0\rangle_e + {}_e\langle 1|\Psi'\rangle_{se}\,\langle\Psi'|1\rangle_e$$

${}_e\langle 0|\Psi'\rangle = \mu\sqrt{n}\,|0\rangle_s + \nu\sqrt{1-n}\,|1\rangle_s \in \mathcal{H}_s$

$$\text{“}\langle\Psi'|0\rangle_e\text{”} = \begin{pmatrix} |\mu|^2 n & \mu\nu^*\sqrt{n}\sqrt{1-n} \\ \mu^*\nu\sqrt{n}\sqrt{1-n} & |\nu|^2(1-n) \end{pmatrix}$$

$$|\psi'\rangle_{SE} = \mu\sqrt{1-\eta}\,|0\rangle_S + \nu\sqrt{\eta}\,|1\rangle_S \in \mathcal{H}_S$$

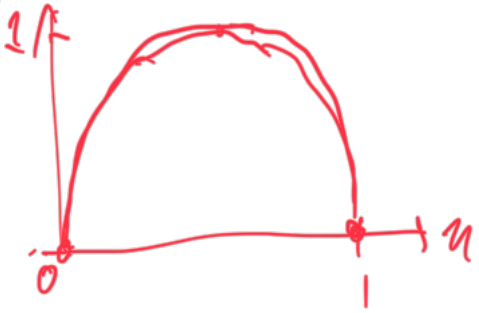
$$\text{“}|\psi'\rangle_E = \begin{bmatrix} |\mu|^2(1-\eta) & \mu\nu^*\sqrt{\eta}\sqrt{1-\eta} \\ \mu^*\nu\sqrt{\eta}\sqrt{1-\eta} & |\nu|^2\eta \end{bmatrix}$$

$$\rho' = \begin{bmatrix} \quad \end{bmatrix} + \begin{bmatrix} \quad \end{bmatrix} = \begin{bmatrix} |\mu|^2 & 2\mu\nu^*\sqrt{\eta}\sqrt{1-\eta} \\ 2\mu^*\nu\sqrt{\eta}\sqrt{1-\eta} & |\nu|^2 \end{bmatrix}$$

$\alpha\,\mu\nu^*$

$= \alpha\,\mu^*\nu$

$$2\sqrt{\eta}\sqrt{1-\eta} = \alpha \le 1$$



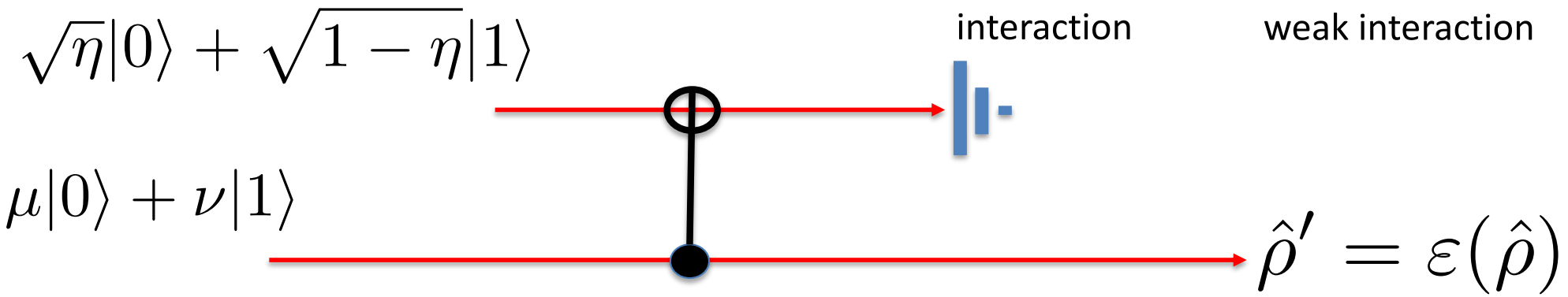
Quantum Decoherence

Example 1: Phase-flip channel

$\eta : 1 \rightarrow \frac{1}{2}$

strong interaction

weak interaction



$$\varepsilon(\hat{\rho}) = |\mu|^2|0\rangle\langle 0| + |\nu|^2|1\rangle\langle 1| + 2\sqrt{\eta(1-\eta)}(\mu\nu^*|0\rangle\langle 1| + \nu\mu^*|1\rangle\langle 0|)$$

decoherence

Quantum Decoherence

Example 1: Phase-flip channel

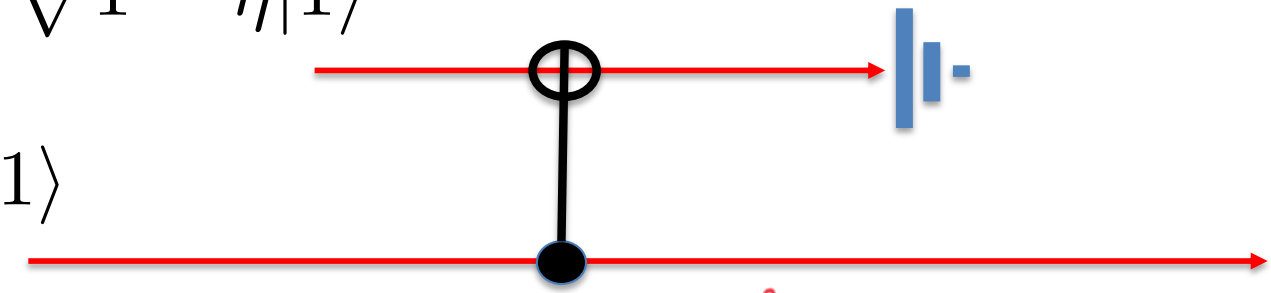
$$\eta: \quad 1 \quad \rightarrow \quad \frac{1}{2}$$

strong interaction

weak interaction

$\sqrt{\eta}|0\rangle + \sqrt{1-\eta}|1\rangle$

$\mu|0\rangle + \nu|1\rangle$



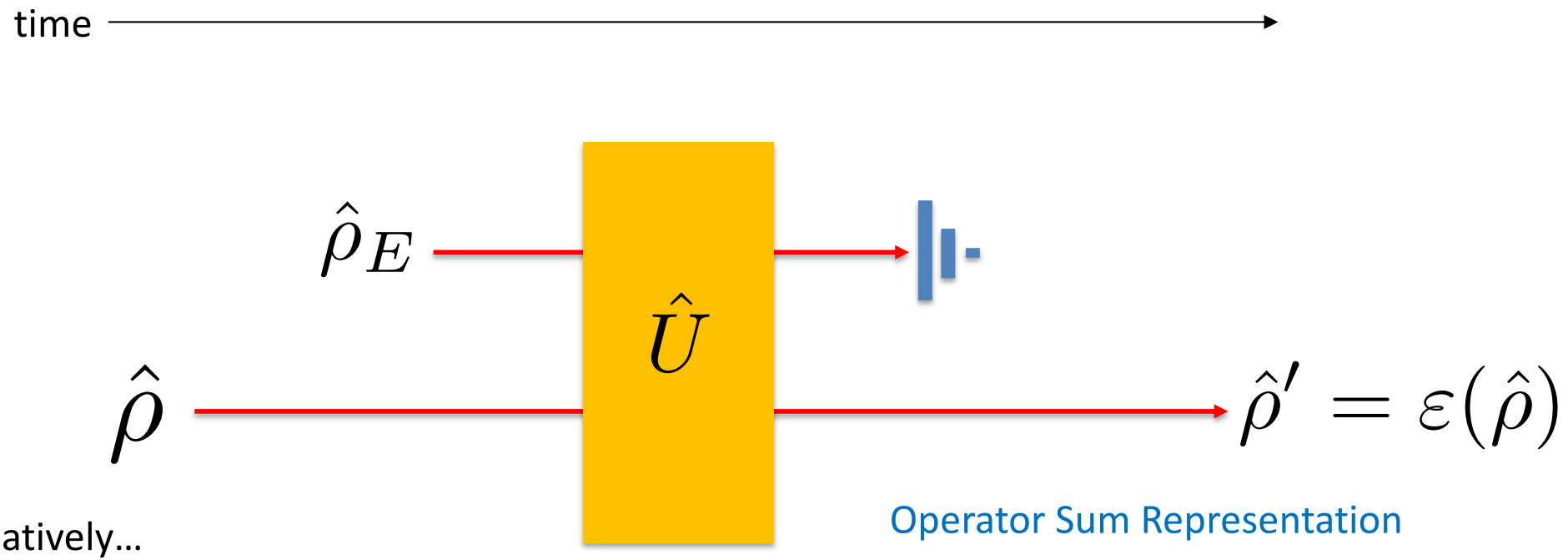
$(-2P$

$$\varepsilon(\hat{\rho}) = |\mu|^2|0\rangle\langle 0| + |\nu|^2|1\rangle\langle 1| + 2\sqrt{\eta(1-\eta)}(\mu\nu^*|0\rangle\langle 1| + \nu\mu^*|1\rangle\langle 0|)$$

decoherence

- Same as phase-flip channel with $P = \frac{1}{2}(1 - 2\sqrt{\eta(1-\eta)})$

Quantum Channels



Alternatively…

Operator Sum Representation

$$\varepsilon(\hat{\rho}) = tr_E[\hat{U}\,\hat{\rho}\otimes\hat{\rho}_E\hat{U}^\dagger] = \sum_k \hat{E}_k\hat{\rho}\hat{E}_k^\dagger$$

where $\hat{E}_k = \langle e_k|\hat{U}|e_0\rangle$ and $\hat{\rho}_E = |e_0\rangle\langle e_0|$

$$\mathcal{E}(\rho_S) = \mathrm{tr}_E\left(U \rho_S \otimes \rho_E U^\dagger\right) = \sum_{e_k} \langle e_k | U(\rho_S \otimes \rho_E) U^\dagger | e_k \rangle$$

m = dimension of the environment.

if $\rho_E = |\psi_0\rangle\langle\psi_0|$ $\Rightarrow \rho_S' = \sum_{k=1}^{m} \langle e_k | U \rho_S \otimes |\psi_0\rangle\langle\psi_0| U^\dagger | e_k \rangle$

$$= \sum_{k} \Big(\underbrace{\sum_e \langle e_k | U_{SE} | \psi_0 \rangle_e}_{\hat{E}_k} \Big) \rho_S \Big(\underbrace{\sum_e \langle \psi_0 | U_{SE}^\dagger | e_k \rangle_e}_{\hat{E}_k^\dagger} \Big)$$

$$= \sum_{k=1}^{m} \underbrace{\hat{E}_k}_{\mathcal{H}_S} \underbrace{\rho_S}_{\mathcal{H}_S} \underbrace{\hat{E}_k^\dagger}_{\mathcal{H}_S}$$

$+ \sum_{k=1}^{m} \hat{E}_k^\dagger \hat{E}_k = \mathbb{1}_S$ because $\mathrm{tr}\,\rho_S' = 1$

$$= \mathrm{tr}\left(\sum_k \hat{E}_k \rho_S \hat{E}_k^\dagger \right) = \sum_k \mathrm{tr}\left(\hat{E}_k \rho_S \hat{E}_k^\dagger \right)$$

$$= \sum_k \mathrm{tr}\left(\hat{E}_k^\dagger \hat{E}_k \rho_S \right) = \mathrm{tr}\left(\left(\sum_k \hat{E}_k^\dagger \hat{E}_k \right) \rho_S \right) \quad \forall \rho_S$$

$$\Rightarrow \sum_k \hat{E}_k^\dagger \hat{E}_k = \mathbb{1}$$

a special case of this operator sum-representation

if $m = 1 \Rightarrow \sum_{k=1}^{1} E_k^\dagger E_k = E_1^\dagger E_1 = \mathbb{1}$

$\Rightarrow$ E_1 is a unitary operator.

So $\rho' = U \rho U^\dagger$ is the operator "sum" rep for unitary evolution

Quantum Channels

Example 1: Phase-flip channel

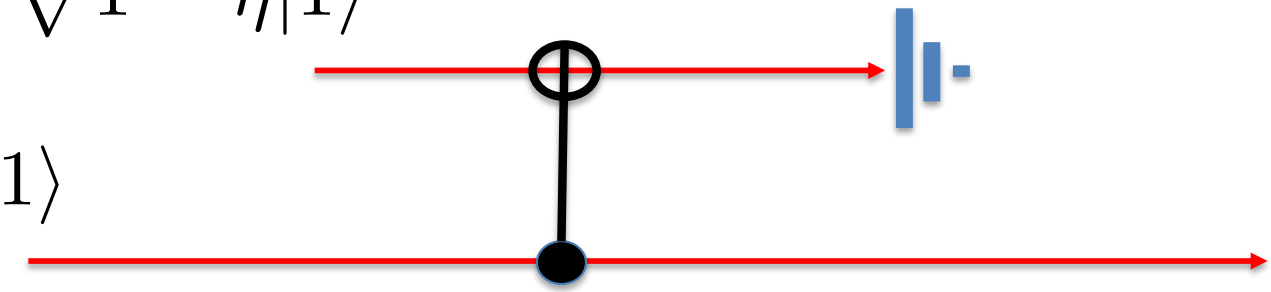
$\eta : 1 \rightarrow \frac{1}{2}$

strong interaction

weak interaction

$\sqrt{\eta}|0\rangle + \sqrt{1-\eta}|1\rangle$

$\mu|0\rangle + \nu|1\rangle$



If we take $\{|e_k\rangle\} = \{|+\rangle, |-\rangle\}$

Operator Sum Representation

then $\{E_k\} = \{\sqrt{1-P}\ \hat{I}, \sqrt{P}\ \hat{Z}\}$ with $P = \frac{1}{2}(1 - 2\sqrt{\eta(1-\eta)})$

so $$\varepsilon(\hat{\rho}) = (1-P)\hat{I}\hat{\rho}\hat{I} + P\hat{Z}\hat{\rho}\hat{Z}$$

- as expected from phase-flip channel

$$|\pm\rangle = \frac{1}{\sqrt{2}}(|0\rangle \pm |1\rangle)$$

$$\mathcal{E}(\rho) = \sum_k \langle e_k | U \rho_S \otimes \rho_E U^\dagger | e_k \rangle$$

$$= \sum_e \langle + | U \rho_S \otimes \rho_E U^\dagger | + \rangle_e + \sum_e \langle - | U \rho_S \otimes \rho_E U^\dagger | - \rangle_e$$

evaluate this last term

recall

$$|\Psi'\rangle_{SE} = \mu\sqrt{n}\,|00\rangle_{SE} + \mu\sqrt{1-n}\,|01\rangle + \nu\sqrt{n}\,|11\rangle + \nu\sqrt{1-n}\,|10\rangle$$

$$\langle - | \Psi' \rangle_E = \frac{1}{\sqrt{2}}\left(\mu\sqrt{n}\,|0\rangle - \mu\sqrt{1-n}\,|0\rangle - \nu\sqrt{n}\,|1\rangle + \nu\sqrt{1-n}\,|1\rangle\right)$$

$$\langle -|0\rangle_E = \frac{1}{\sqrt{2}}$$
$$\langle -|1\rangle = -\frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{n} - \sqrt{1-n}}{\sqrt{2}}\left(\mu|0\rangle_S - \nu|1\rangle_S\right)$$

$$= \frac{\sqrt{n} - \sqrt{1-n}}{\sqrt{2}}\, Z\,|\Psi_S\rangle$$

$$= \left(\frac{\sqrt{n} - \sqrt{1-n}}{\sqrt{2}}\right)^2 Z\,|\Psi\rangle_S\langle\Psi|\,Z^\dagger = \left(\frac{\sqrt{n} - \sqrt{1-n}}{\sqrt{2}}\right)^2 Z\rho_S Z^\dagger$$

$$= \underbrace{\frac{1}{2}\left(1 - 2\sqrt{n}\sqrt{1-n}\right)}_{= p} Z\rho_S Z^\dagger$$

$$= \left(\frac{\sqrt{x} + \sqrt{1-x}}{\sqrt{2}} \right)^2 \mathbb{1} \rho_S \mathbb{1} = \frac{1}{2} \underbrace{\left(1 + 2\sqrt{x}\sqrt{1-x}\right)}_{1-p} \mathbb{1} \rho_S \mathbb{1}$$

$$\mathcal{E}(\rho) = (1-p)\ \mathbb{1} \rho \mathbb{1} + p\ Z \rho Z^\dagger$$

$$\hat{E}_0 = \sqrt{1-p}\,\mathbb{1} \qquad \hat{E}_1 = \sqrt{p}\,Z$$

$$\text{so} \quad \mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger$$

Quantum Channels

Example 1: Phase-flip channel

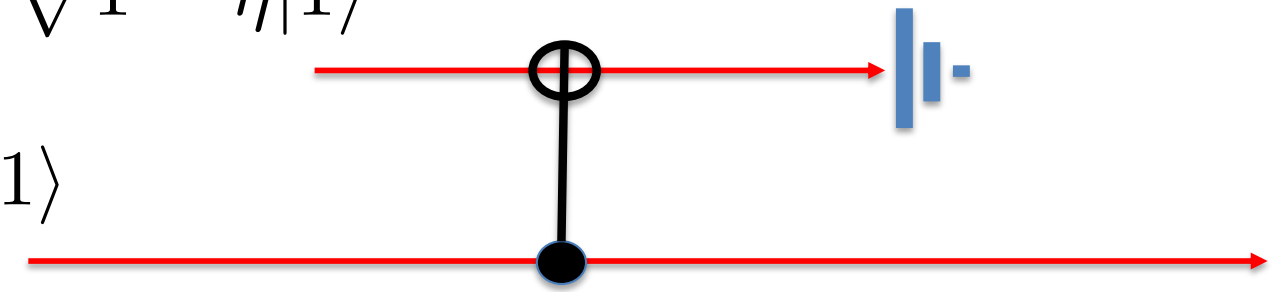
$\eta : \quad 1 \quad \rightarrow \quad \frac{1}{2}$

strong interaction

weak interaction

$\sqrt{\eta}|0\rangle + \sqrt{1-\eta}|1\rangle$

$\mu|0\rangle + \nu|1\rangle$



If we take $\{|e_k\rangle\} = \{|0\rangle, |1\rangle\}$

Operator Sum Representation

then

$$\{E_k\} = \{\sqrt{\eta}|0\rangle\langle 0| + \sqrt{1-\eta}|1\rangle\langle 1|, \sqrt{\eta}|1\rangle\langle 1| + \sqrt{1-\eta}|0\rangle\langle 0|\}$$

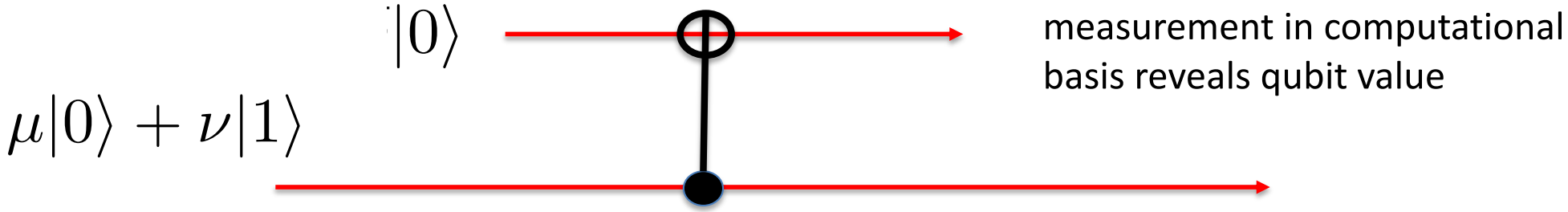
This has the same effect as $\{\sqrt{1-p}\,\mathbb{1}, \sqrt{p}\,Z\}$

- as expected from a partial measurement channel

Quantum Channels

Example 1(b): Projective measurement channel $\eta: 1$

strong interaction



If we take $\{|e_k\rangle\} = \{|0\rangle, |1\rangle\}$

Operator Sum Representation

then $\{E_k\} = \{|0\rangle\langle 0|, |1\rangle\langle 1|\}$

$$\varepsilon(\hat{\rho}) = |\mu|^2|0\rangle\langle 0| + |\nu|^2|1\rangle\langle 1|$$

completely mixed state: coherences / off-diagonal terms → 0

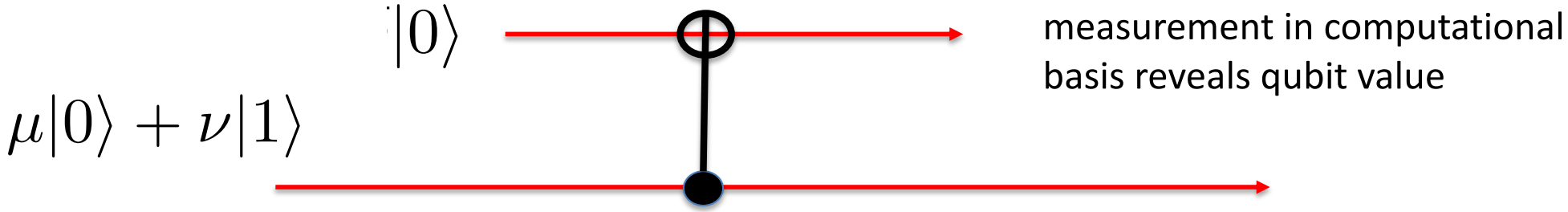
- as expected from a projective measurement channel

Quantum Decoherence

Example 1: Phase-flip channel

$\eta : \quad 1$

strong interaction



measurement in computational basis reveals qubit value

If we take $\{|e_k\rangle\} = \{|0\rangle, |1\rangle\}$

then $\{E_k\} = \{|0\rangle\langle 0|, |1\rangle\langle 1|\}$

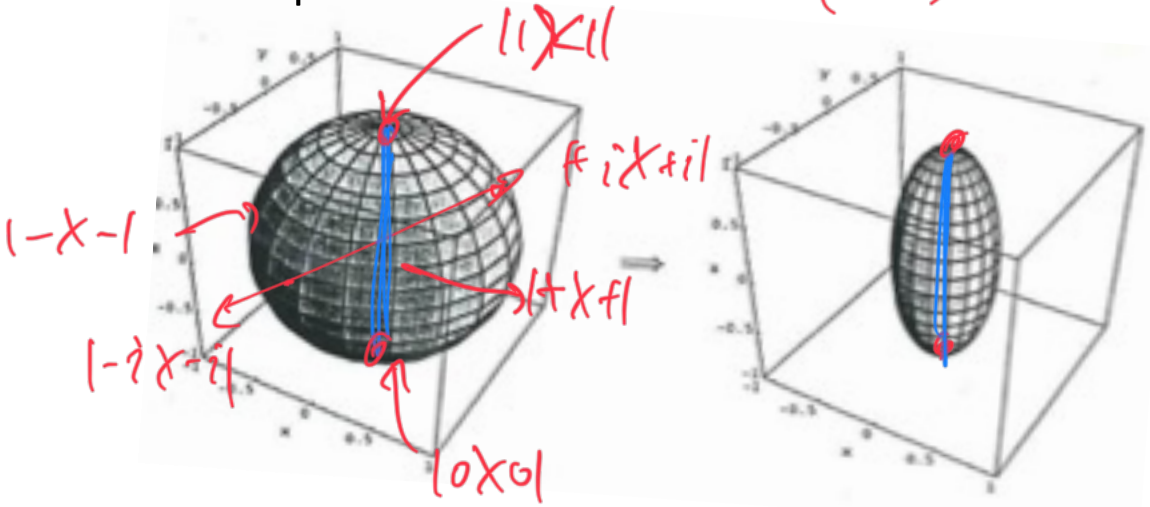
50:50 phase flip channel = strongly decohering channel = projective measurement channel = **classical channel**

$$\varepsilon(\hat{\rho}) = |\mu|^2|0\rangle\langle 0| + |\nu|^2|1\rangle\langle 1|$$ ← **fixed point of the channel**

- as expected from a projective measurement channel

Quantum Decoherence

Example 1: Phase-flip channel



If we take $\{|e_k\rangle\} = \{|+\rangle, |-\rangle\}$

Operator Sum Representation

then $\{E_k\} = \{\sqrt{1-P}\,\hat{I}, \sqrt{P}\,\hat{Z}\}$ with $P = \frac{1}{2}(1 - 2\sqrt{\eta(1-\eta)})$

so $$\varepsilon(\hat{\rho}) = (1-P)\hat{I}\hat{\rho}\hat{I} + P\hat{Z}\hat{\rho}\hat{Z}$$

- as expected from phase-flip channel

Bloch vector

$$\rho \rightarrow \underline{r} = (r_x, r_y, r_z) = \left(\mathrm{tr}(\rho\hat{x}), \mathrm{tr}(\rho\hat{y}), \mathrm{tr}(\rho\hat{z})\right)$$

if $\rho = |\psi\rangle\langle\psi| \Rightarrow \underline{r} = \left(\langle\psi|\hat{x}|\psi\rangle, \langle\psi|\hat{y}|\psi\rangle, \langle\psi|\hat{z}|\psi\rangle\right)$

(pure state)

$+ \quad |\underline{r}|^2 = \underline{1}$ (surface of the Bloch sphere)

$\rho = \underline{1}/2$ (completely mixed):

$\underline{r}_{1/2} = (0, 0, 0)$ Bloch vector

Quantum Channels

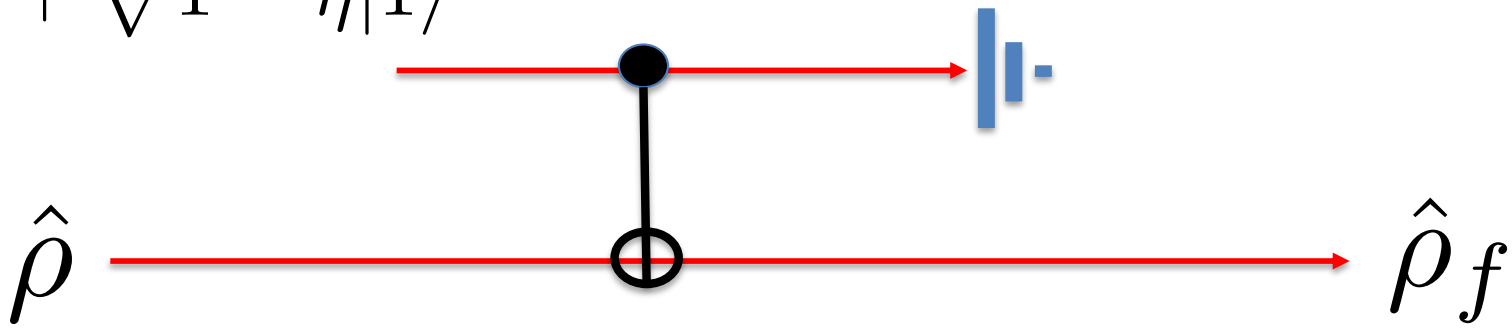
Example 2: Bit-flip channel

$\eta: \quad 1 \quad \rightarrow \quad \frac{1}{2}$

weak interaction

strong interaction

$\sqrt{\eta}|0\rangle + \sqrt{1-\eta}|1\rangle$



If we take $\{|e_k\rangle\} = \{|0\rangle, |1\rangle\}$

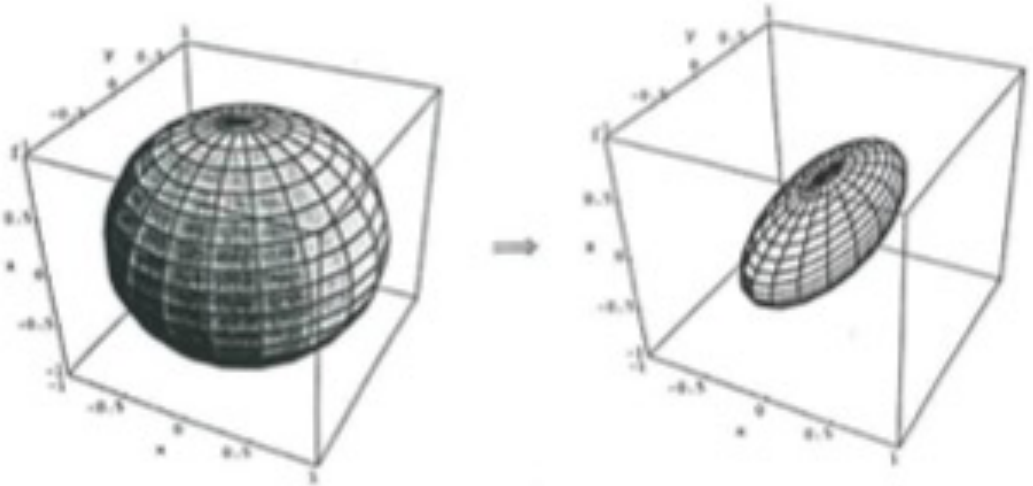
Operator Sum Representation

then $\{E_k\} = \{\sqrt{1-P}\ \hat{I}, \sqrt{P}\ \hat{X}\}$ with $P = 1-\eta$

so $$\varepsilon(\hat{\rho}) = (1-P)\hat{I}\hat{\rho}\hat{I} + P\hat{X}\hat{\rho}\hat{X}$$

Quantum Channels

Example 2: Bit-flip channel



If we take $\{|e_k\rangle\} = \{|0\rangle, |1\rangle\}$

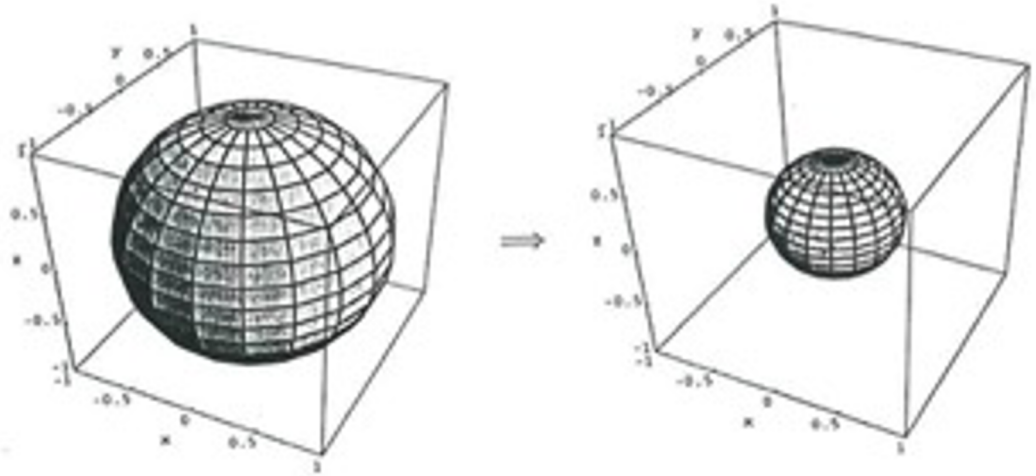
Operator Sum Representation

then $\{E_k\} = \{\sqrt{1-P}\,\hat{I}, \sqrt{P}\,\hat{X}\}$ with $P = 1 - \eta$

so $$\varepsilon(\hat{\rho}) = (1-P)\hat{I}\hat{\rho}\hat{I} + P\hat{X}\hat{\rho}\hat{X}$$

Quantum Channels

Example 3: Depolarizing channel



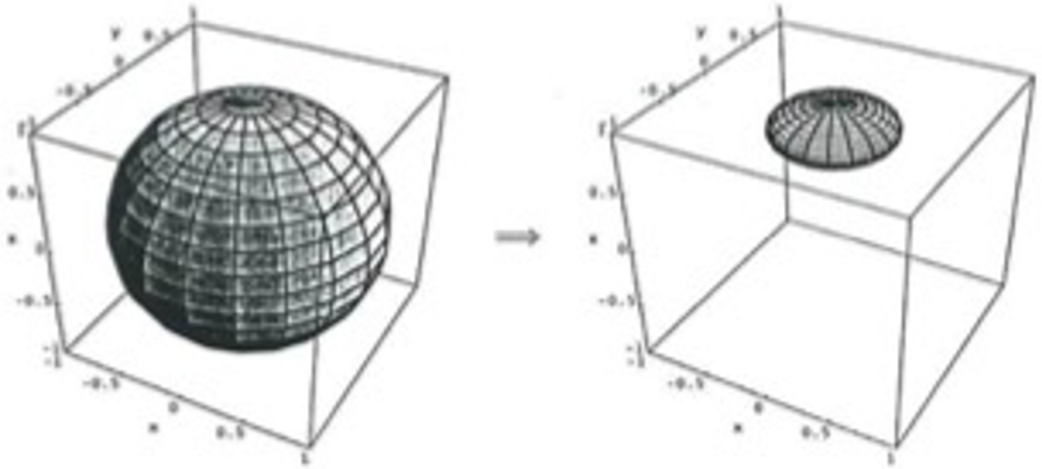
$$\varepsilon(\hat{\rho}) = (1 - \frac{3P}{4})\rho + \frac{P}{4}(\hat{X}\hat{\rho}\hat{X} + \hat{Y}\hat{\rho}\hat{Y} + \hat{Z}\hat{\rho}\hat{Z})$$

information is degraded in all bases

$$\frac{1}{2}\hat{I}$$ ← Fixed point is maximally mixed

Quantum Channels

Example 4: Amplitude damping channel



$$\varepsilon(\hat{\rho}) = \sum_k \hat{E}_k \hat{\rho} \hat{E}_k^\dagger$$

where $\{E_k\} = \{|0\rangle\langle 0| + \sqrt{1-\gamma}|1\rangle\langle 1|, \sqrt{\gamma}|0\rangle\langle 1|\}$

and γ is probability of losing a quanta of energy to environment

$|0\rangle\langle 0|$ ← Fixed point is the ground state

Quantum Channels

Example 4(a): Generalized Amplitude damping channel

$$\varepsilon(\hat{\rho}) = \sum_k \hat{E}_k \hat{\rho} \hat{E}_k^\dagger$$

where

$$\{E_k\} = \{\sqrt{P}(|0\rangle\langle 0| + \sqrt{1-\gamma}|1\rangle\langle 1|), \sqrt{1-P}(|1\rangle\langle 1| + \sqrt{1-\gamma}|0\rangle\langle 0|),$$
$$\sqrt{P}\sqrt{\eta}|0\rangle\langle 1|, \sqrt{1-P}\sqrt{\eta}|1\rangle\langle 0|\}$$

and γ is probability of losing or gaining a quanta of energy from environment

$$P|0\rangle\langle 0| + (1-P)|1\rangle\langle 1|$$ ← Fixed point

Quantum Channels

Example 4(a): Generalized Amplitude damping channel

$$\varepsilon(\hat{\rho}) = \sum_k \hat{E}_k \hat{\rho} \hat{E}_k^\dagger$$

where

$$\{E_k\} = \{\sqrt{P}(|0\rangle\langle 0| + \sqrt{1-\gamma}|1\rangle\langle 1|), \sqrt{1-P}(|1\rangle\langle 1| + \sqrt{1-\gamma}|0\rangle\langle 0|), \sqrt{P}\sqrt{\eta}|0\rangle\langle 1|, \sqrt{1-P}\sqrt{\eta}|1\rangle\langle 0|\}$$

if $$P = \frac{e^{-E_0/(k_B T)}}{e^{-E_0/(k_B T)} + e^{-E_1/(k_B T)}}$$

and γ is probability of losing or gaining a quanta of energy from environment

$$P|0\rangle\langle 0| + (1-P)|1\rangle\langle 1|$$ ← Fixed point is thermal equilibrium

Quantum Channels

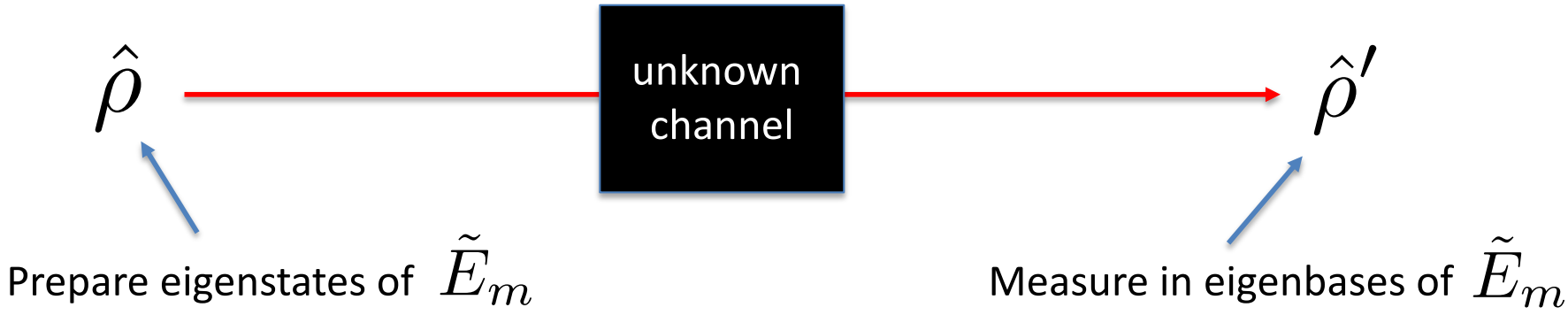
Chi matrix representation

$$\varepsilon(\hat{\rho}) = \sum_k \hat{E}_k \hat{\rho} \hat{E}_k^\dagger = \sum_{m,n} \tilde{E}_m \hat{\rho} \tilde{E}_n^\dagger \, \chi_{m,n}$$

matrix which completely characterizes the channel

where the $\tilde{E}_m$ are a complete set of basis operators

e.g: $\{\tilde{E}_m\} = \{\hat{I}, \hat{X}, -i\hat{Y}, \hat{Z}\}$



determine the elements of $\chi_{m,n}$

Summary

- quantum channels can model open system evolution via a quantum circuit interaction with an environmental state
  - explicit models of decoherence effects
  - explicit models of energy decay effects
- operator sum representation is a powerful approach
  - different physical processes lead to the same channel
- allows "black-box" characterization of channels

Discrete evolution input state → output states

Next : continuous evolution with noise → Lindblad